B.Sc. (Hons.) in Environmental Science

- B.Sc. (Hons.) in Environmental Science is a bachelor's degree program that aims to provide students with a comprehensive understanding of the natural environment and the human impacts on it .
- The program covers a wide range of topics, such as ecology, geology, hydrology, atmospheric science, conservation biology, and environmental management .
- The program also involves practical, hands-on learning through fieldwork, laboratory experiments, and projects that expose students to the skills and technology used in environmental science .
- The program prepares students for careers in various sectors, such as environmental consultancy, conservation, education, research, policy, and industry .
- The program typically lasts for three or four years, depending on the institution and the country .
- The program may have different entry requirements, such as academic qualifications, English proficiency, and personal statement, depending on the institution and the country .
- The program may also offer opportunities for students to study abroad, take a placement year, or pursue a dual degree, depending on the institution and the country .

Unit 1 - Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness. Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.

Environment: Definition, Types of Environment, Components of environment, Segments of environment, Scope and importance, Need for Public Awareness.

- Environment is the sum total of all the physical, chemical and biological factors and processes that surround and influence a living organism.
- There are two types of environment: natural and human-made. Natural environment consists of the natural resources and phenomena that exist independently of human intervention, such as air, water, soil, plants, animals, climate, etc. Human-made environment consists of the artificial structures and systems that humans create and modify, such as buildings, roads, dams, industries, farms, etc.
- The components of the environment are mainly divided into two categories: biotic and abiotic. Biotic components are the living organisms that interact with each other and with the abiotic components. Abiotic components are the non-living physical and chemical factors that affect the biotic components, such as temperature, light, moisture, pH, etc.
- The environment can be divided into four segments or spheres based on the three components of the environment: atmosphere, hydrosphere, lithosphere and biosphere. Atmosphere is the layer of gases that surrounds the earth and protects it from the harmful radiation and meteorites from outer space. Hydrosphere is the water component of the earth, which includes oceans, rivers, lakes, glaciers, etc. Lithosphere is the solid component of the earth, which includes rocks, minerals, soils, etc. Biosphere is the zone of life, where the biotic and abiotic components interact with each other .
- The environment is important for the survival and well-being of all living organisms, as it provides the essential resources and services, such as oxygen, water, food, shelter, climate regulation, nutrient cycling, etc. The environment also influences the evolution, adaptation, behavior and

distribution of living organisms .

- The environment is facing various threats and challenges due to the increasing human population, consumption and activities, such as deforestation, pollution, climate change, biodiversity loss, resource depletion, etc. These threats and challenges pose serious risks to the health, security and sustainability of the environment and the living organisms .
- There is a need for public awareness and education about the environment, its importance, its problems and its solutions. Public awareness and education can help to create a sense of responsibility and concern among the people, and motivate them to adopt environmentally friendly practices and policies, such as conservation, recycling, renewable energy, etc. Public awareness and education can also help to foster a positive attitude and appreciation towards the environment and its diversity .

Ecosystem: Definition, Types of ecosystem, Structure of ecosystem, Food Chain, Food Web, Ecological pyramid. Balance Ecosystem. Effects of Human Activities such as Food, Shelter, Housing, Agriculture, Industry, Mining, Transportation, Economic and Social security on Environment, Environmental Impact Assessment, Sustainable Development.

- Ecosystem is a functional unit of the environment, where a group of living organisms interact with each other and with the abiotic components in a specific area.
- There are two types of ecosystem: natural and human-made. Natural ecosystem is a self-regulating and self-sustaining system that is formed by natural processes, such as forest, grassland, desert, etc. Human-made ecosystem is a modified or artificial system that is created or influenced by human activities, such as crop field, urban area, aquarium, etc .
- The structure of an ecosystem consists of two components: biotic and abiotic. The biotic component is further divided into three levels: producers, consumers and decomposers. Producers are the autotrophic organisms that can make their own food by using light or chemical energy, such as plants, algae, etc. Consumers are the heterotrophic organisms that depend on other organisms for food, such as animals, fungi, etc. Decomposers are the organisms that break down the dead or waste organic matter into simpler substances, such

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Unit 2 - Natural Resources: Introduction, Classification

- Natural resources are substances or materials from nature that humans use to sustain life. These resources are naturally available on Earth without any human action .
- Natural resources have economic, ecological, scientific, aesthetic, and cultural values in human life.
- Natural resources can be classified into different categories based on their origin, level of development and uses, stock or deposits, and their distribution.
- On the basis of their origin, natural resources can be classified into living or biotic and non-living or abiotic resources.
  - Living or biotic resources are derived from living things or organic materials, such as plants, animals, microorganisms, etc .
  - Non-living or abiotic resources are derived from nonliving or inorganic materials, such as minerals, metals, fossil fuels, water, air, etc .
- On the basis of their level of development and uses, natural resources can be classified into actual and potential resources.
  - Actual resources are those that have been surveyed, quantified, and are being used in the present.
  - Potential resources are those that have been identified and estimated, but are not being used in the present due to lack of technology, accessibility, or demand.
- On the basis of their stock or deposits, natural resources can be classified into renewable and non-renewable resources.
  - Renewable resources are those that can be regenerated or replenished by natural processes within a reasonable time span, such as sunlight, wind, water, biomass, etc .
  - Non-renewable resources are those that have a finite supply and cannot be regenerated or replenished by natural processes within a reasonable time span, such as coal, oil, natural gas, metals, etc .
- On the basis of their distribution, natural resources can be classified into ubiquitous and localized resources.
  - Ubiquitous resources are those that are found everywhere or in most places on Earth, such as air, sunlight, water, etc.
  - Localized resources are those that are found only in certain regions or places on Earth, such as petroleum, coal, iron ore, etc.

Water Resources; Availability, sources and Quality Aspects, Water Borne and Water Induced Diseases, Fluoride and Arsenic Problems in Drinking Water

- Water is one of the most essential and abundant natural resources on Earth. It covers about 71% of the Earth's surface and constitutes about 60% of the human body.

- Water is a renewable resource, but its availability and quality are affected by various factors, such as climate change, population growth, pollution, overexploitation, etc.
- Water resources can be classified into surface water and groundwater sources.
  - Surface water sources include rivers, lakes, ponds, streams, wetlands, etc. that are directly exposed to the atmosphere.
  - Groundwater sources include aquifers, springs, wells, etc. that are located below the Earth's surface and are recharged by the infiltration of rainwater or surface water.
- Water quality refers to the physical, chemical, and biological characteristics of water that determine its suitability for various purposes, such as drinking, irrigation, industrial, recreational, etc.
- Water quality is influenced by various factors, such as natural processes, human activities, land use, climate, etc.
- Water quality can be measured by various parameters, such as pH, dissolved oxygen, turbidity, temperature, hardness, salinity, nutrients, metals, pathogens, etc.
- Water quality standards are the legal or regulatory limits that specify the acceptable levels of various parameters in water for different uses.
- Water quality standards vary from country to country and depend on the intended use of water, the availability of water, the level of development, the environmental and health concerns, etc.
- Water borne diseases are the diseases that are caused by the ingestion or contact with contaminated water that contains harmful microorganisms, such as bacteria, viruses, protozoa, etc.
- Some examples of water borne diseases are cholera, typh

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution and their Effects

- Pollution is the introduction of harmful substances or energy into the environment that cause adverse effects on living and non-living things.
- Pollution can be in the form of substances (solid, liquid, gaseous) or energy (sound, heat, radiation, etc.).
- Pollution affects the natural balance and creates adverse conditions for life. It has led to severe environmental issues that require immediate action and resolution.
- Some of the major effects of pollution are:
  - Environment degradation: Pollution can reduce the quality of air,

  water, and soil, leading to smog, acid rain, eutrophication, soil erosion, desertification, etc. These can affect the biodiversity, climate, and natural resources of the environment.
  - Human health: Pollution can cause various diseases and disorders in humans, such as respiratory infections, asthma, allergies, cancer, cardiovascular problems, neurological damage, etc. Pollution can also affect the mental and reproductive health of humans .
  - Economic and social impacts: Pollution can reduce the productivity and efficiency of various sectors, such as agriculture, fisheries, tourism, industry, etc. Pollution can also increase the costs of health care, environmental remediation, and disaster management. Pollution can also affect the social and cultural values of people, such as their quality of life, well-being, and happiness.

Public Health Aspects of Environmental

- Public health is the science and practice of protecting and improving the health of populations through the prevention and control of diseases, injuries, and other health problems.
- Environmental factors are one of the determinants of public health, as they can influence the exposure, susceptibility, and response of people to various health hazards.
- Some of the public health aspects of environmental are:
  - Environmental health: This is the branch of public health that deals with the assessment, management, and communication of environmental risks to human health. It covers topics such as environmental epidemiology, toxicology, exposure assessment, risk assessment, environmental standards, and regulations.
  - Environmental justice: This is the principle that all people have the right to live in a healthy and safe environment, regardless of their race, ethnicity, income, or other social factors. It also implies that all people have the right to participate in the decision-making processes that affect their environment and health.
  - Environmental sustainability: This is the concept that the current generation should meet their needs without compromising the ability of future generations to meet their own needs. It also implies that the human activities should respect the limits and capacities of the natural environment and its resources.

Water Pollution

- Water pollution is the contamination of water bodies, such as rivers, lakes, oceans, groundwater, etc., by harmful substances or energy that impair their quality and usability.
- Water pollution can be caused by various sources, such as domestic sewage, industrial effluents, agricultural runoff, urban stormwater, oil spills, ra-

dioactive waste, thermal pollution, etc.

- Water pollution can have various effects, such as:
  - Ecological impacts: Water pollution can affect the aquatic ecosystems, such as the diversity, abundance, and functioning of the organisms, the food webs, the nutrient cycles, the oxygen levels, etc. Water pollution can also cause the loss or degradation of habitats, such as wetlands, coral reefs, mangroves, etc.
  - Human health impacts: Water pollution can cause various diseases and infections in humans, such as diarrhea, cholera, typhoid, hepatitis, dysentery, etc. Water pollution can also cause chronic effects, such as cancer, neurological disorders, endocrine disruption, etc. Water pollution can also affect the quality and safety of drinking water, recreational water, and seafood.
  - Economic and social impacts: Water pollution can affect various sectors, such as agriculture, fisheries, tourism, industry, etc., by reducing their productivity, quality, and profitability. Water pollution can also increase the costs of water treatment, health care, and environmental restoration. Water pollution can also affect the cultural and aesthetic values of water bodies, such as their religious, historical, and recreational significance.

Air Pollution

- Air pollution is the presence of harmful substances or energy in the atmosphere that cause adverse effects on living and non-living things[^11

Unit 4 - Current Environmental Issues of Importance

- The world is facing many environmental problems that threaten the well-being of humans and other living beings. Some of these problems are global in scale, while others are local or regional. Some of these problems are caused by human activities, while others are natural phenomena. Some of these problems are urgent and require immediate action, while others are long-term and require sustained efforts.

- Some of the current environmental issues of importance are:

  - **Global warming**: This is the rise in the average temperature of the Earth's surface and atmosphere due to the increased concentration of greenhouse gases (such as carbon dioxide, methane, nitrous oxide, etc.) in the air. These gases trap the heat from the sun and prevent it from escaping back into space, creating a greenhouse effect. Global warming causes many adverse impacts, such as melting of ice caps and glaciers, sea level rise, extreme weather events, droughts, floods, wildfires, heat waves, crop failures, biodiversity loss, diseases, conflicts, and migration.

- **Greenhouse effect**: This is the natural process by which the Earth's atmosphere absorbs and re-emits some of the infrared radiation from the sun, keeping the planet warm and habitable. Without the greenhouse effect, the Earth would be too cold for life to exist. However, human activities have enhanced the greenhouse effect by adding more greenhouse gases to the atmosphere, making the Earth warmer than it would be otherwise.

- **Climate change**: This is the long-term change in the statistical distribution of weather patterns over periods of time ranging from decades to millions of years. Climate change can be caused by natural factors, such as changes in the Earth's orbit, solar activity, volcanic eruptions, and plate tectonics, or by human factors, such as greenhouse gas emissions, land use change, aerosol pollution, and deforestation. Climate change affects the physical, biological, and social systems of the Earth, such as the water cycle, the carbon cycle, the ecosystems, the agriculture, the health, the economy, and the security.

- **Acid rain**: This is the precipitation that contains high levels of acidic substances, such as sulfur dioxide and nitrogen oxides, which are emitted by the burning of fossil fuels, such as coal, oil, and gas. Acid rain can lower the pH of soil and water, damaging the plants, animals, and buildings that depend on them. Acid rain can also cause respiratory problems, skin irritation, and eye damage in humans and other animals.

- **Ozone layer formation and depletion**: This is the process by which the ozone molecules (O3) are created and destroyed in the stratosphere, the layer of the atmosphere that lies between 10 and 50 kilometers above the Earth's surface. Ozone is formed by the reaction of oxygen molecules (O2) with ultraviolet (UV) radiation from the sun, and is destroyed by the reaction of ozone molecules with other chemicals, such as chlorine and bromine, which are released by human activities, such as the use of chlorofluorocarbons (CFCs), halons, and methyl bromide. The ozone layer protects the Earth from the harmful effects of UV radiation, such as skin cancer, eye cataracts, immune system suppression, and crop damage. The depletion of the ozone layer allows more UV radiation to reach the Earth's surface, increasing the risks of these effects.

- **Population growth and automobile pollution**: This is the increase in the number of people living on the Earth and the increase in the number of vehicles that they use for transportation. Population growth and automobile pollution are interrelated, as more people demand more mobility, more energy, and more resources, leading to more emissions of greenhouse gases and other pollutants, such as carbon monoxide, nitrogen oxides, sulfur dioxide, particulate matter, and volatile organic compounds. These emissions contribute to global warming, greenhouse effect, climate change, acid rain, ozone depletion, smog, and respiratory diseases.

  - **Burning of paddy straw**: This is the practice of setting fire to the leftover stalks of rice plants after harvesting, which is done by some farmers in India and other countries to clear the fields for the next crop. Burning of paddy straw causes air pollution, as it releases

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act 1986 is an Act of the Parliament of India that aims to provide for the protection and improvement of the environment.
- The Act was enacted in May 1986 and came into force on 19 November 1986. It has 26 sections and 4 chapters.
- The Act is widely considered to have been a response to the Bhopal gas leak, one of the worst industrial disasters in history, that occurred in December 1984.
- The Act empowers the Central Government to establish authorities, issue rules and directions, and take measures to prevent, control and abate environmental pollution.
- The Act also provides for penalties and punishments for violating its provisions or rules, as well as for handling hazardous substances.

Initiatives by Non Governmental Organizations (NGO's) for environmental protection

- Non Governmental Organizations (NGOs) are non-profit organizations that are citizen-led and unaffiliated with the government. They play an important role in raising environmental awareness, advocating for environmental causes, and implementing environmental projects and programs.
- Some of the initiatives taken by NGOs for environmental protection are:
  - Environmental education and awareness among people, especially children and youth, about the importance of conserving natural resources, reducing waste, and adopting sustainable lifestyles.
  - Environmental pollution control, such as monitoring air, water, soil, and noise quality, promoting renewable energy sources, and campaigning against polluting industries and practices.

- Protection of forest wealth, such as preventing deforestation, promoting afforestation and social forestry, and supporting indigenous and local communities that depend on forests for their livelihoods and culture.
- Wildlife conservation, such as protecting endangered species and habitats, combating poaching and illegal trade, and promoting ecotourism and wildlife-friendly agriculture.
- Recycling and waste utilization, such as promoting the 3Rs (reduce, reuse, recycle), developing waste management systems, and converting waste into useful products and energy.
- Rural development and eco development, such as empowering rural communities, especially women and marginalized groups, to participate in decision-making and benefit from natural resources, and promoting organic farming, watershed management, and biodiversity conservation.

Human Population and the Environment

- The human population has experienced a period of unprecedented growth, more than tripling in size since 1950. It reached almost 7.8 billion in 2020 and is projected to grow to over 8.5 billion in 2030 and 9.7 billion in 2050.
- The ways in which populations are spread across Earth has an effect on the environment. Developing countries tend to have higher birth rates due to poverty and lower access to family planning and education, while developed countries have lower birth rates but higher consumption and emission levels per capita.
- The impact of so many humans on the environment takes two major forms:
  - Consumption of resources such as land, food, water, air, fossil fuels, and minerals, which leads to depletion, degradation, and pollution of natural resources and ecosystems.
  - Waste products as a result of consumption such as air and water pollutants, toxic materials, and greenhouse gases, which contribute to global warming, climate change, and health problems.
- Population growth, environmental degradation, and climate change are interrelated and mutually reinforcing challenges that require integrated and holistic solutions. Some of the possible solutions are:
  - Education, especially for women and girls, which can improve health, income, and empowerment, and reduce fertility and mortality rates.
  - Environmental education, which can increase awareness and knowledge of environmental issues and promote pro-environmental attitudes and behaviors.
  - Population-environmental education, which can link human geography, demography, and ecology, and address the complex and diverse

relationships between population and environment at multiple levels and periods of analysis.

- Sustainable development, which can balance the economic, social, and environmental dimensions of human well-being, and ensure that the needs of the present generation are met without compromising the ability of future generations to meet their own needs.